GRR on the universal curve: six details

27 October 2026

Reference. D. Zvonkine, *An introduction to moduli spaces of curves and their intersection theory*, §3; page numbers are the printed ones. The sections follow the order of the reference. We assume $2g-2+n>0$, work on the smooth Deligne–Mumford stacks, and use rational cohomology. The universal curve is $\pi : \mathcal{C} = \overline{\mathcal{C}}_{g,n} \to \overline{\mathcal{M}}_{g,n}$; other notation is local to each section.

1 The pushforward $p_!$ in K-theory (§3.1.4, p. 696)

Notation. Zvonkine reserves p_* for the cohomological pushforward (fiber integration) and writes $R^k(p,V)$ for the sheaves below. We use the standard notation $R^k p_* V$. The sheaf $R^0 p_* V = p_* V$ has nothing to do with $p_* : H^*(X) \to H^*(Y)$.

Precise definition. Let $p : X \to Y$ be proper, with X, Y smooth, and let $\mathcal{F}$ be a coherent sheaf on X, for instance the sheaf of sections of a vector bundle V. The higher direct image $R^k p_* \mathcal{F}$ is the sheaf on Y associated to the presheaf $U \mapsto H^k(p^{-1}(U), \mathcal{F})$, i.e. the kth right derived functor of p_*. It is coherent (Grothendieck; Grauert in the analytic setting) and vanishes when k exceeds the fiber dimension. Define

$$\boxed{p_![\mathcal{F}] = \sum_{k\geq 0} (-1)^k \, [R^k p_* \mathcal{F}] \in K^0(Y).}$$

The classes $[R^k p_* \mathcal{F}]$ lie in $K^0(Y)$ because coherent sheaves on Y have finite locally free resolutions (the remark in the middle of p. 696). This uses algebraicity: it holds on smooth quasi-projective varieties and on smooth quotient stacks such as $\overline{\mathcal{M}}_{g,n}$. A short exact sequence of sheaves gives a *finite* long exact sequence of the $R^k p_*$, so $p_!$ is additive and descends to $K^0(X)$. In derived language, $p_![\mathcal{F}]$ is the class of the complex $Rp_*\mathcal{F}$. Three cases occur in these notes:

- $Y = \mathrm{pt}$: $p_! V = \chi(X,V) \in K^0(\mathrm{pt}) = \mathbb{Z}$ (Example 3.7).
- A closed embedding i: i_* is exact, so $i_![\mathcal{F}] = [i_*\mathcal{F}]$. This gives the class of $i_*\mathcal{O}_\Delta$ in Section 4. It is a sheaf, not a vector bundle, and the Koszul resolution exhibits its class in K^0.
- The universal curve: $\pi_! L = [\pi_* L] - [R^1\pi_* L] = \Lambda - \mathbb{C}$ (Proposition 3.11).

The descriptive definition (p. 696, bottom). Zvonkine says that the spaces $H^k(X_y, V)$, $y \in Y$, "form a sheaf" $R^k(p,V)$. The precise relation is the *base change map*

$$\varphi_y^k : \; R^k p_* V \otimes k(y) \longrightarrow H^k(X_y, V|_{X_y})$$

from the fiber of the sheaf to the cohomology of the fiber. It need not be an isomorphism. Assume that p is flat (true for π), V is locally free, and Y is reduced. The following holds (Hartshorne, *Algebraic Geometry*, III.12.9 and III.12.11; Mumford, *Abelian Varieties*, §5).

1. $y \mapsto \chi(X_y, V)$ is locally constant. Each $h^k(X_y, V)$ is only upper semicontinuous: it can jump up on closed subsets.
2. (Grauert) If $h^k(X_y, V)$ is locally constant, then $R^k p_* V$ is a vector bundle and every φ_y^k is an isomorphism. In this case Zvonkine's description is literally correct.
3. In general, $Rp_* V$ is represented by a complex of vector bundles $K^0 \to K^1 \to \cdots \to K^d$ on Y. This holds locally on Y by Mumford §5, and globally on the spaces of this course. Its cohomology sheaves are the $R^k p_* V$, and $H^k(K^\bullet \otimes k(y)) \cong H^k(X_y, V)$ for *every* y. Hence

$$p_! V = \sum_k (-1)^k [K^k].$$

Item 3 is the honest form of "$p_!V = H^0 - H^1 + \cdots$ fiberwise." The groups $H^k(X_y, V)$ may jump, but they are the fiberwise cohomology of one complex of vector bundles. Only the alternating sum of its terms is well defined, as a class in $K^0(Y)$ of rank $\chi(X_y, V)$.

The universal curve. The proof of Proposition 3.11 (pp. 699–700) uses item 2. Every stable curve C has $h^0(C, \omega_C) = g$ and, by Serre duality, $h^1(C, \omega_C) = 1$. So $\Lambda = \pi_*\omega_\pi$ is a vector bundle of rank g with fiber $H^0(C, \omega_C)$. Likewise, $R^1\pi_*\omega_\pi$ is a line bundle with fiber $H^1(C, \omega_C)$, trivialized by the trace map.

An example of jumping. Let E be an elliptic curve with origin 0, let $p = \mathrm{pr}_2 : X = E \times E \to E$, and let $P = \mathcal{O}_X(\Delta_E - \{0\} \times E - E \times \{0\})$, so that $P|_{E\times\{y\}} \cong \mathcal{O}_E(y - 0)$. Then $h^0 = h^1 = 0$ for $y \neq 0$, and $h^0 = h^1 = 1$ at $y = 0$. The sheaf p_*P is torsion free of generic rank 0, so $p_*P = 0$, although $H^0(X_0, P) = \mathbb{C}$; thus φ_0^0 is not surjective. The sheaf R^1p_*P is the skyscraper $\mathcal{O}_0$ (Mumford §13), which is not a vector bundle. A complex as in item 3 is $\mathcal{O}_E(-0) \xrightarrow{s} \mathcal{O}_E$, where s is the canonical section. On fibers s is zero at $y = 0$ and an isomorphism elsewhere, which matches the jump. Hence $p_!P = [\mathcal{O}_E(-0)] - [\mathcal{O}_E] = -[\mathcal{O}_0]$. GRR confirms this. Here $\mathrm{Td}(p) = \mathrm{pr}_1^* \mathrm{Td}(TE) = 1$, and $c_1(P)^2 = -2[\mathrm{pt}]$ because $\Delta_E^2 = 0$ and the three curves meet pairwise once. Since P has degree 0 on the fibers, only $\mathrm{ch}_2(P)$ contributes, and $\mathrm{ch}(p_!P) = p_*(c_1(P)^2/2) = -[\mathrm{pt}] = -\mathrm{ch}(\mathcal{O}_0)$.

2 Dividing by a Todd class (Theorem 3.6, p. 697)

Let $p : X \to Y$ be a proper morphism of complex manifolds, $d = \dim X$. Zvonkine sets $\mathrm{Td}(p) = \mathrm{Td}(TX)/\mathrm{Td}(p^*TY)$. No localization is needed to divide: $\mathrm{Td}(p^*TY) = 1 + u$ with u of positive degree, so $u^{d+1} = 0$ and

$$\mathrm{Td}(p^*TY)^{-1} = 1 - u + u^2 - \cdots + (-u)^d.$$

Relation to the usual GRR. The familiar form is $\mathrm{ch}(p_!V)\,\mathrm{Td}(TY) = p_*\big(\mathrm{ch}(V)\,\mathrm{Td}(TX)\big)$, with $p_!$ as in Section 1. Multiply by $\mathrm{Td}(TY)^{-1}$ and move it inside p_* with the projection formula $p_*(\alpha)\,\beta = p_*(\alpha\, p^*\beta)$; since $p^*\,\mathrm{Td}(TY) = \mathrm{Td}(p^*TY)$,

$$\mathrm{ch}(p_!V) = p_*\big(\mathrm{ch}(V)\ \mathrm{Td}(TX)\ \mathrm{Td}(p^*TY)^{-1}\big) = p_*\big(\mathrm{ch}(V)\,\mathrm{Td}(p)\big),$$

which is Theorem 3.6. The inverse must be pulled back because the integrand is a class on X.

Meaning. The Todd class is multiplicative, so $\mathrm{Td}(p)$ is the Todd class of the virtual relative tangent bundle $[TX] - [p^*TY] \in K^0(X)$. If p is a submersion, the sequence $0 \to T_{X/Y} \to TX \to p^*TY \to 0$ gives $\mathrm{Td}(p) = \mathrm{Td}(T_{X/Y})$. The universal curve is not a submersion at the nodes, so there is no relative tangent bundle. The quotient is still defined, but computing it requires a correction supported on the node locus (pp. 700–702, Sections 3–4 below). GRR applies to π on the stack because its fibers are honest curves (p. 699).

3 The Koszul resolution in rank two (Theorem 3.10, pp. 698–699)

Setup. Let $q : V \to X$ be a holomorphic vector bundle of rank two (Zvonkine's "$p = 2$" is the rank). All sheaves live on the *total space* V, and $\mathcal{O}_X$ means functions on the zero section extended by zero to V. Put $\mathcal{V} = q^*V$ and $A^k = \bigwedge^k \mathcal{V}^\vee$, so $A^0 = \mathcal{O}_V$.

The differential. The bundle $\mathcal{V}$ has a tautological section $\tau(b, v) = v \in V_b = \mathcal{V}_{(b,v)}$, and $d : A^k \to A^{k-1}$ is contraction with it: $(d\alpha)_{(b,v)}(w_1, \ldots) = \alpha_{(b,v)}(v, w_1, \ldots)$. This map is $\mathcal{O}_V$-linear; it is *not* an exterior derivative. Alternation gives $d \circ d = 0$. In coordinates t on X and fiber coordinates (x, y) for a local trivialization, let dx, dy be the dual frame of the fiber directions (there are no dt_j terms). Then $\tau = xe_1 + ye_2$ and

$$d(dx) = x, \qquad d(dy) = y, \qquad d(dx \wedge dy) = x\,dy - y\,dx.$$

With Zvonkine's labels, where (1) and (5) are the zero maps, the sequence is

$$0 \xrightarrow{(1)} A^2 \xrightarrow{(2)} A^1 \xrightarrow{(3)} \mathcal{O}_V \xrightarrow{(4)} \mathcal{O}_X \xrightarrow{(5)} 0,$$

(2) $f\,dx \wedge dy \mapsto -yf\,dx + xf\,dy$, (3) $g_x\,dx + g_y\,dy \mapsto xg_x + yg_y$, (4) $h(t;x,y) \mapsto h(t;0,0)$.

The subscripts on g_x, g_y are labels, not derivatives. Contraction and restriction are intrinsic, so these local formulas glue to global sheaf maps.

Exactness. It suffices to check stalks. At a point of the zero section, the stalk R consists of convergent power series in $(t;x,y)$, and the complex reads $0 \to R \to R^2 \to R \to R/(x,y) \to 0$. Two facts are used: R and $R/(x) = \mathbb{C}\{t,y\}$ are integral domains; and *(Div)* if $g(t;0,y) = 0$ then $g = xf$ with $f \in R$, since every monomial of g contains x.

- *At A^2.* If $xf = yf = 0$ then $f = 0$, because R is a domain.
- *At A^1 (the main step).* Let $xg_x + yg_y = 0$. Setting $x = 0$ gives $y\,g_y(t;0,y) = 0$ in $\mathbb{C}\{t,y\}$, so $g_y(t;0,y) = 0$ and $g_y = xf$ by (Div). Then $x(g_x + yf) = 0$, so $g_x = -yf$, and $g_x\,dx + g_y\,dy = d(f\,dx \wedge dy)$. Conversely, $d \circ d = 0$.
- *At $\mathcal{O}_V$.* The image of (3) is the ideal (x,y), which restricts to zero. Conversely, if $h(t;0,0) = 0$, then $h = [h(t;x,y) - h(t;0,y)] + [h(t;0,y) - h(t;0,0)] = xa + yb$ by (Div) applied twice.
- *At $\mathcal{O}_X$.* A germ $a(t)$ is the restriction of $h(t;x,y) = a(t)$.

Off the zero section the stalk of $\mathcal{O}_X$ vanishes and x or y is a unit. If x is a unit, then $h = d\big((h/x)\,dx\big)$ and $f = g_y/x$ works at A^1; the case of y is symmetric.

Reading the reference. Zvonkine's "$f = -g_x/y = g_y/x$" means holomorphic division *after* divisibility has been proved, as above; it is valid even where $x = y = 0$. "The first map" in that paragraph is map (2). Although the matrices of d vanish at $x = y = 0$, the stalk maps do not: the theorem asserts exactness of sheaves, not of fibers.

4 The node locus and "$\Delta = \nu_1\nu_2$" (pp. 691, 701–702)

Setup. Let $i : \Delta \hookrightarrow \mathcal{C}$ be the locus of nodes. Locally $\mathcal{C}$ is $xy = t$ over the base and Δ is $x = y = 0$, so Δ is smooth of codimension two. Let $N = N_{\Delta/\mathcal{C}}$, $Y = \mathrm{Tot}(N)$, with projection $q : Y \to \Delta$ and zero section $s : \Delta \to Y$. On p. 702 the total space is that of N; the dual $N^\vee$ enters only as a term of the Koszul resolution of Section 3. With pullbacks displayed, it reads

$$0 \to \textstyle\bigwedge^2 q^*N^\vee \to q^*N^\vee \to \mathcal{O}_Y \to s_*\mathcal{O}_\Delta \to 0.$$

The symbols ν_1, ν_2 (p. 691). They are the formal Chern roots of $N^\vee$: $\nu_1 + \nu_2 = -c_1(N)$ and $\nu_1\nu_2 = c_2(N)$. On the double cover of Δ that orders the two branches, they become the first Chern classes of the two branch cotangent lines. A symmetric polynomial $P(\nu_1,\nu_2)$ is a class on Δ, and Zvonkine's $\Delta \cdot P(\nu_1,\nu_2)$ means the Gysin pushforward

$$\Delta \cdot P(\nu_1,\nu_2) := i_*P(\nu_1,\nu_2) \in H^{*+4}(\mathcal{C}).$$

The self-intersection and projection formulas give

$$i^*i_*\alpha = c_2(N)\,\alpha, \qquad (i_*\alpha)(i_*\beta) = i_*\big(c_2(N)\,\alpha\beta\big), \qquad [\Delta]^2 = i_*c_2(N).$$

So "identifying $\nu_1\nu_2$ with Δ" is the rule $\nu_1\nu_2\,P \mapsto i_*P$. It is not an equality in one ring: $c_2(N)$ lives on Δ, while $[\Delta] = i_*1$ lives on $\mathcal{C}$.

Zvonkine's computation (p. 702). Let $T(z) = z/(1-e^{-z})$, so $\mathrm{Td}(L) = T(c_1(L))$ for a line bundle L. The Koszul resolution and multiplicativity give, on Y,

$$\mathrm{Td}(s_*\mathcal{O}_\Delta)^{-1} = \frac{\mathrm{Td}(q^*N^\vee)}{\mathrm{Td}(\bigwedge^2 q^*N^\vee)} = q^*A, \qquad A = \frac{T(\nu_1)T(\nu_2)}{T(\nu_1+\nu_2)}.$$

Since $T(0)=1$, $A-1$ vanishes when $\nu_1=0$ or $\nu_2=0$, so $A = 1+\nu_1\nu_2 F$ for a symmetric power series F. Zvonkine's algebra gives

$$F = \sum_{k\ge 1} \frac{B_{2k}}{(2k)!}\,\frac{\nu_1^{2k-1}+\nu_2^{2k-1}}{\nu_1+\nu_2} = \frac{1}{12} - \frac{\nu_1^2-\nu_1\nu_2+\nu_2^2}{720} + \cdots,$$

where each quotient is a polynomial because its numerator vanishes at $\nu_2=-\nu_1$. Replacing $\nu_1\nu_2$ by Δ yields the following.

Proposition.

$$\boxed{\mathrm{Td}(i_*\mathcal{O}_\Delta)^{-1} = 1 + i_*F(\nu_1,\nu_2) = 1 + \frac{[\Delta]}{12} - \frac{1}{720}\,i_*\big(c_1(N)^2 - 3c_2(N)\big) + \cdots}$$

Why this needs an argument. Zvonkine moves the calculation from Y to $\mathcal{C}$ because characteristic classes are topological: a neighborhood U of Δ in $\mathcal{C}$ is diffeomorphic to a neighborhood of the zero section in Y. Restricting to U alone loses information, however: the class of a point on a compact complex surface is nonzero but restricts to zero on a small ball. What does transfer are classes *with supports in* Δ. The Koszul complex is exact off the zero section, so it defines a class in $K^0_\Delta(Y)$, and excision gives $H^*_\Delta(Y) \cong H^*_\Delta(\mathcal{C})$. Under this isomorphism, $i_*\alpha$ is the image of $q^*\alpha \cdot u_N$, where u_N is the Thom class.[1] The first proof below avoids this machinery. It uses only GRR for i and the self-intersection formula. The second proof carries the machinery out and does not use GRR.

First proof (GRR). Put $E = i_*\mathcal{O}_\Delta$. Formal quotients by $\nu_1\nu_2$ below are taken in the ring of symmetric power series before evaluating on Δ; no cohomology class is inverted.

Step 1: Chern character. For the embedding i, $\mathrm{Td}(i) = \mathrm{Td}(T\Delta)/\mathrm{Td}(i^*T\mathcal{C}) = \mathrm{Td}(N)^{-1}$, so Theorem 3.6 gives $\mathrm{ch}(E) = i_*\,\mathrm{Td}(N)^{-1}$. The roots of N are $-\nu_1, -\nu_2$ and $T(-\nu)^{-1} = (e^\nu-1)/\nu$. Hence

$$\mathrm{Td}(N)^{-1} = \frac{(e^{\nu_1}-1)(e^{\nu_2}-1)}{\nu_1\nu_2} = \frac{e^{\nu_1+\nu_2}-e^{\nu_1}-e^{\nu_2}+1}{\nu_1\nu_2}.$$

Since i_* raises degree by four, we get $\mathrm{ch}_0(E) = \mathrm{ch}_1(E) = 0$ and, for $m \ge 2$,

$$\mathrm{ch}_m(E) = i_*\left(\frac{(\nu_1+\nu_2)^m - \nu_1^m - \nu_2^m}{m!\,\nu_1\nu_2}\right).$$

Step 2: logarithm. Write $\ell(z) = \log T(z) = \sum_{m\ge1} a_m z^m$. For a bundle with roots x_j, $\log \mathrm{Td} = \sum_j \ell(x_j) = \sum_m a_m\, m!\,\mathrm{ch}_m$. Both sides are additive, so the identity holds on K^0, hence for E, which has a finite locally free resolution. Step 1 gives

$$\log \mathrm{Td}(E) = -i_*R, \qquad R = \frac{\ell(\nu_1)+\ell(\nu_2)-\ell(\nu_1+\nu_2)}{\nu_1\nu_2}, \qquad \text{so} \qquad \nu_1\nu_2 R = \log A.$$

Step 3: exponentiate. By induction on the product formula above, $(i_*R)^r = i_*\big((\nu_1\nu_2)^{r-1}R^r\big)$ for $r \ge 1$. Therefore

$$\begin{aligned}\mathrm{Td}(E)^{-1} = \exp(i_*R) &= 1 + i_*\left(\sum_{r\ge1}\frac{(\nu_1\nu_2)^{r-1}R^r}{r!}\right) = 1 + i_*\left(\frac{\exp(\nu_1\nu_2 R)-1}{\nu_1\nu_2}\right)\\ &= 1 + i_*\left(\frac{A-1}{\nu_1\nu_2}\right) = 1 + i_*F.\end{aligned}$$

[1] For complexes exact off a closed subset, see P. Baum, W. Fulton, R. MacPherson, *Riemann–Roch and topological K-theory for singular varieties*, §§1.4–1.5 and §2, pp. 162–168.

The low-degree terms use $\nu_1^2 - \nu_1\nu_2 + \nu_2^2 = (\nu_1+\nu_2)^2 - 3\nu_1\nu_2$. □

Second proof (Thom classes and Gysin maps, no GRR). This is Zvonkine's route: compute on the normal bundle, then transfer through the tubular neighbourhood. The transfer is made honest by working on the Thom space of N, which carries exactly the classes supported near Δ. To keep the topology classical, we write the argument for a compact complex manifold X, projective so that coherent sheaves have finite locally free resolutions, and a closed complex submanifold Δ of codimension two. N, Y, q, s and $E = i_*\mathcal{O}_\Delta$ are as above. For $X = \mathcal{C}$ the same steps run in orbifold topology with rational coefficients, just as the first proof applies GRR on the stack; we do not discuss this.

Step 0: the Gysin map through the collapse. Fix a hermitian metric on N, and let $D(N) \subset Y$ and $S(N)$ be the closed unit disc bundle and the unit sphere bundle. The Thom space is $\mathrm{Th}(N) = D(N)/S(N)$, so $\widetilde{H}^*(\mathrm{Th}\,N) = H^*(D(N), S(N))$ and $\widetilde{K}^0(\mathrm{Th}\,N) = K^0(D(N), S(N))$. By the tubular neighbourhood theorem, $D(N)$ maps diffeomorphically onto a closed neighbourhood D of Δ in X. The map is the identity on Δ and induces the identity on normal bundles; it is not holomorphic. Collapsing $X \setminus \operatorname{int} D$ to a point gives the Pontryagin–Thom map

$$c : X \longrightarrow X/(X \setminus \operatorname{int} D) \cong D/\partial D \cong \mathrm{Th}(N).$$

Let $u \in \widetilde{H}^4(\mathrm{Th}\,N)$ be the Thom class of the complex bundle N. The Thom isomorphism $\Phi(\alpha) = q^*\alpha \cdot u$ maps $H^k(\Delta)$ isomorphically onto $\widetilde{H}^{k+4}(\mathrm{Th}\,N)$, and

$$i_* = c^* \circ \Phi : \; H^k(\Delta) \longrightarrow H^{k+4}(X)$$

is the topological definition of the Gysin map. It agrees with the Poincaré duality definition (Bott–Tu, *Differential Forms in Algebraic Topology*, §6). In forms: multiply $q^*\alpha$ by a Thom form supported in D and extend by zero. Forgetting supports, u restricts to q^*e on $D(N)$, where $e = c_2(N) = \nu_1\nu_2$ is the Euler class. Hence

$$\Phi(\alpha)\,\Phi(\beta) = q^*(\alpha\beta)\,u \cdot u = q^*(e\,\alpha\beta)\,u = \Phi(e\,\alpha\beta). \tag{$*$}$$

So $\widetilde{H}^*(\mathrm{Th}\,N)$ is $H^*(\Delta)$ with its product twisted by e. This is the rule "$\nu_1\nu_2 P \mapsto i_*P$" on the Thom space, and applying c^* gives the product formula for i_* stated above.

Step 1: from sheaves to topology. A finite complex $G_\bullet$ of vector bundles on a compact space Z, exact at every point of a closed subset A, defines a class $[G_\bullet] \in K^0(Z, A)$ (the difference construction; see the footnote above). The class maps to $\sum_k (-1)^k [G_k]$ in $K^0(Z)$. It is natural for maps of pairs, unchanged by homotopies through complexes exact over A, and zero if $G_\bullet$ is exact on all of Z. The Koszul complex of Section 3, restricted to $D(N)$, is exact over $S(N)$. It defines the *K-theoretic Thom class*

$$\theta = \big[0 \to \textstyle\bigwedge^2 q^*N^\vee \to q^*N^\vee \to \mathcal{O} \to 0\big] \in K^0(D(N), S(N)) = \widetilde{K}^0(\mathrm{Th}\,N).$$

Localization lemma. $[E] = c^*\theta$ in $K^0(X)$.

This makes precise Zvonkine's "characteristic classes are topological". It is the only step that uses holomorphic geometry; in the first proof, GRR for i plays this role. The proof is given at the end of this section.

Step 2: reduction to the Thom space. The class θ vanishes at the base point, so $\mathrm{Td}(\theta)^{-1} = 1 + \varepsilon$ with $\varepsilon \in \widetilde{H}^*(\mathrm{Th}\,N)$. By the Thom isomorphism, $\varepsilon = \Phi(G)$ for a unique $G \in H^*(\Delta)$. Characteristic classes are natural under the continuous map c, so the lemma gives

$$\mathrm{Td}(E)^{-1} = c^*\,\mathrm{Td}(\theta)^{-1} = 1 + c^*\Phi(G) = 1 + i_*G.$$

It remains to show $G = F$, a statement about N alone. Forgetting supports, i.e. restricting to $D(N) \simeq \Delta$, recovers Zvonkine's computation on Y: θ restricts to $q^*[\bigwedge^\bullet N^\vee]$, so $\varepsilon \mapsto A - 1$, while

$\Phi(G) \mapsto eG$. Thus $eG = A - 1 = eF$. This does *not* give $G = F$ when e is a zero divisor in $H^*(\Delta)$; for instance $e = 0$ if N has a nowhere vanishing section. This is the restriction problem described above, now in one line. We repair it by placing $D(N)$ inside a compact space on which forgetting supports is injective.

Step 3: the projective completion. Let $p : P = \mathbb{P}(N \oplus \mathbb{C}) \to \Delta$ be the bundle of lines in $N \oplus \mathbb{C}$, write points as $[v : \lambda]$, and let $P_\infty = \{\lambda = 0\} = \mathbb{P}(N)$. The open set $\{\lambda \neq 0\} = \{[v : 1]\}$ is Y, so $D(N) \subset Y \subset P$; the zero section is $\Delta_0 = \{[0 : 1]\}$. Let $\mathcal{O}(-1) \subset p^*(N \oplus \mathbb{C})$ be the tautological line bundle, and put

$$Q = p^*N \otimes \mathcal{O}(1) = \mathrm{Hom}(\mathcal{O}(-1), p^*N), \qquad \sigma :\ \mathcal{O}(-1) \hookrightarrow p^*(N \oplus \mathbb{C}) \xrightarrow{\mathrm{pr}} p^*N.$$

At $[v : \lambda]$ the section σ sends $(v, \lambda) \mapsto v$, so it vanishes exactly on Δ_0. On Y the vector $(v, 1)$ trivializes $\mathcal{O}(-1)$. This identifies $Q|_Y = q^*N$, and $\sigma|_Y$ becomes the tautological section $\tau(v) = v$ of Section 3; in particular $Q|_{\Delta_0} = N$. So on the compact manifold P, the zero section is cut out by a section of a rank-two bundle extending q^*N. Here "$\nu_1\nu_2 = \Delta$" will be literally true. Let $c_P : P \to \mathrm{Th}(N)$ be the collapse map for $D(N) \subset P$.

(a) c_P^* *is injective on* $\widetilde{H}^*(\mathrm{Th}\, N)$. After excision, c_P^* is the map $H^*(P, P \setminus \mathrm{int}\, D(N)) \to H^*(P)$. Its kernel is the image of the connecting map from $H^{*-1}(P \setminus \mathrm{int}\, D(N))$. The scaling $[v : \lambda] \mapsto [v : t\lambda]$, $t \in [0, 1]$, deformation retracts $P \setminus \mathrm{int}\, D(N) = \{|v| \geq |\lambda|\}$ onto P_∞. By Leray–Hirsch, $H^*(P_\infty)$ is generated over $H^*(\Delta)$ by $c_1(\mathcal{O}(1))$, which is restricted from P. So $H^*(P) \to H^*(P_\infty)$ is surjective and the connecting map vanishes.

(b) $c_P^*\theta = 1 - [Q^\vee] + [\bigwedge^2 Q^\vee]$ *in* $K^0(P)$. The Koszul complex $\bigwedge^2 Q^\vee \to Q^\vee \to \mathcal{O}_P$ (contraction with σ) is defined on all of P and is exact off Δ_0. On $D(N)$ it is the complex defining θ. By naturality, its class in $K^0(P, P \setminus \mathrm{int}\, D(N))$ corresponds to θ under excision, and its image in $K^0(P)$ is the alternating sum.

(c) $c_P^*u = c_2(Q)$. The Euler class is $c_2(Q) = \sigma^*u_Q$, where u_Q is the Thom class of Q on its total space; any section is homotopic to the zero section. Since $\sigma \neq 0$ on $P \setminus \mathrm{int}\, D(N)$, the class σ^*u_Q comes from $H^4(P, P \setminus \mathrm{int}\, D(N)) \cong H^4(D(N), S(N))$. There it equals $\sigma^*u_{q^*N}$, where $\sigma(v) = (v, v)$ in $\mathrm{Tot}(q^*N) = \{(v, w) : q(v) = q(w)\}$. The bundle map $(v, w) \mapsto w$ to $\mathrm{Tot}(N)$ pulls u back to u_{q^*N}, and its composite with σ is the identity of $D(N)$. Hence $\sigma^*u_{q^*N} = u$, and $c_P^*u = c_2(Q)$. This is the familiar fact that the Euler class is dual to the transverse zero locus of a section: $c_2(Q) = [\Delta_0]$.

Now let F_Q be F evaluated on the Chern roots of $Q^\vee$. It is a class on P, and $F_Q|_{D(N)} = q^*F(\nu_1, \nu_2)$ because $Q|_{D(N)} = q^*N$. So $\Phi(F) = F_Q|_{D(N)} \cdot u$, and by naturality of the cup product and (c),

$$c_P^*\Phi(F) = F_Q \cdot c_P^*u = c_2(Q)\, F_Q.$$

On the other hand, use (b), multiplicativity, and the identity $A = 1 + \nu_1\nu_2 F$ for the roots ν_1, ν_2 of $Q^\vee$, for which $\nu_1\nu_2 = c_2(Q^\vee) = c_2(Q)$. This gives

$$c_P^* \,\mathrm{Td}(\theta)^{-1} = \mathrm{Td}\big(1 - Q^\vee + \textstyle\bigwedge^2 Q^\vee\big)^{-1} = \dfrac{\mathrm{Td}(Q^\vee)}{\mathrm{Td}(\bigwedge^2 Q^\vee)} = 1 + c_2(Q)\, F_Q.$$

Hence $c_P^*\varepsilon = c_P^*\Phi(F)$, and (a) gives $\varepsilon = \Phi(F)$, i.e. $G = F$. By Step 2, $\mathrm{Td}(E)^{-1} = 1 + i_*F$. □

Remarks. (1) On P the slogan is an honest equation in one cohomology ring: $\nu_1\nu_2$ becomes $c_2(Q)$, which is the class of Δ_0. Zvonkine's computation on Y is correct, but it has to be read on the Thom space, whose products obey $(*)$. Step 3 shows that nothing is lost there. (2) Replace Td^{-1} by ch in Steps 2–3. Then $c_P^* \,\mathrm{ch}(\theta) = \prod_j (1 - e^{\nu_j}) = c_2(Q)\,\mathrm{Td}(Q)^{-1}$, so $\mathrm{ch}(\theta) = \Phi(\mathrm{Td}(N)^{-1})$ and $\mathrm{ch}(E) = i_*\,\mathrm{Td}(N)^{-1}$. This is Step 1 of the first proof. So the lemma also yields the case of GRR used there.

Proof of the localization lemma. For a complex of vector bundles exact off Δ, write $[\,\cdot\,]_D$ for its class in $K^0(D,\partial D)\cong K^0(D(N),S(N))$. The map c^* is the composite $K^0(D,\partial D)\cong K^0(X,X\setminus \operatorname{int} D)\to K^0(X)$, and $[E]=\sum_k(-1)^k[F_k]$ for *any* finite locally free resolution $F_\bullet\to E$. So it suffices to find one such resolution with $[F_\bullet]_D=\theta$.

(i) Independence of the resolution. Let $G_\bullet$ and $G'_\bullet$ be finite locally free resolutions of the same sheaf S on a smooth quasi-projective variety, with S supported on a closed set Z. Then, on every compact pair (W,B) with $B\cap Z=\emptyset$, they have the same class in $K^0(W,B)$. Since there are enough vector bundles, there is a third finite resolution $R_\bullet\to S$ with chain maps $f:R_\bullet\to G_\bullet$ and $f':R_\bullet\to G'_\bullet$ over id_S. Build R_k inductively as a vector bundle surjecting onto the relevant fibre product. Stop after dim steps, when the kernel is locally free because the variety is smooth. The cone of the quasi-isomorphism f is a bounded exact complex of vector bundles, so it is exact on fibres and its class is 0. Replace f by tf, $t\in[0,1]$. Each cone is exact off Z, being a cone of a map between complexes exact there, and at $t=0$ the cone is $G_\bullet\oplus R_\bullet[1]$. Hence $0=[G_\bullet]-[R_\bullet]$, and likewise $[G'_\bullet]=[R_\bullet]$.

(ii) Deformation to the normal bundle (Fulton, *Intersection Theory*, Ch. 5). Let M be the blow-up of $X\times\mathbb{C}$ along $\Delta\times 0$, minus the strict transform of $X\times 0$, and let $\rho:M\to\mathbb{C}$ be the projection. Then M is smooth and quasi-projective, ρ is a submersion, $\rho^{-1}(\mathbb{C}^*)=X\times\mathbb{C}^*$, and $\rho^{-1}(0)=Y$. The strict transform $\widetilde{\Delta}\cong\Delta\times\mathbb{C}$ of $\Delta\times\mathbb{C}$ meets $\rho^{-1}(t)$ in Δ for $t\neq 0$, and in the zero section of Y for $t=0$. In local coordinates with $X=\mathbb{C}^2_x\times\mathbb{C}^m_z$ and $\Delta=\{x=0\}$, we have $M=\mathbb{C}^2_w\times\mathbb{C}^m_z\times\mathbb{C}_t$ with blow-down map $(w,z,t)\mapsto(tw,z,t)$, and $\widetilde{\Delta}=\{w=0\}$. At $t=0$ the coordinate w is the normal vector $w\,\partial_x\in N$. So ∂_w identifies $N_{\widetilde{\Delta}/M}$ with the pullback of N, by the identity at $t=0$ and at $t=1$.

Choose a finite locally free resolution $G_\bullet\to\mathcal{O}_{\widetilde{\Delta}}$ on M. All its terms are flat over $\mathbb{C}$, so it stays exact on each fibre of ρ. Thus $F_\bullet:=G|_{t=1}$ resolves E on X, and $G|_{t=0}$ resolves $s_*\mathcal{O}_\Delta$ on Y. Over the segment $[0,1]\subset\mathbb{C}$, choose a closed tubular neighbourhood $\mathcal{D}\cong D(N)\times[0,1]$ of $\widetilde{\Delta}\cap\rho^{-1}[0,1]$ inside the fibres of ρ, using the exponential map of a metric on $\ker d\rho$. Since tubular neighbourhoods are unique up to isotopy, we may take its slices to be D at $t=1$ and the standard $D(N)$ at $t=0$. Now $G_\bullet$ is exact on $\mathcal{D}\setminus\widetilde{\Delta}$, so homotopy invariance gives $[F_\bullet]_D=[G|_{t=0}]_{D(N)}$ in $K^0(D(N),S(N))$. By (i) on Y, the right side is the class of the Koszul resolution, which is θ. $\square$

5 Chern classes from the Chern character (Proposition 3.17, Corollary 3.19, pp. 703–704)

The general identity. For every complex vector bundle E, with $\mathrm{ch}_m(E)\in H^{2m}$ normalized as in Definition 3.3 (p. 694),

$$\boxed{c(E)=\exp\Big(\sum_{m\geq 1}(-1)^{m-1}(m-1)!\;\mathrm{ch}_m(E)\Big).}$$

Positive-degree classes are nilpotent, so exp and log are finite sums. *Proof.* By the splitting principle (Proposition 3.4, p. 695) we may assume that E has Chern roots $x_1,\dots,x_r$. Then $\sum_j x_j^m=m!\;\mathrm{ch}_m(E)$, and

$$\log c(E)=\sum_j\log(1+x_j)=\sum_{m\geq1}\frac{(-1)^{m-1}}{m}\sum_j x_j^m=\sum_{m\geq1}(-1)^{m-1}(m-1)!\;\mathrm{ch}_m(E).\qquad\square$$

Correction to the printed statement. The exponent is $\mathrm{ch}_1-\mathrm{ch}_2+2!\;\mathrm{ch}_3-3!\;\mathrm{ch}_4+\cdots$, so the factorials *multiply*. The divisions printed in Proposition 3.17 and in the proof of Corollary 3.19 are typographical errors. Expanding gives

$$c_1=\mathrm{ch}_1,\qquad c_2=\tfrac12\,\mathrm{ch}_1^2-\mathrm{ch}_2,\qquad c_3=\tfrac16\,\mathrm{ch}_1^3-\mathrm{ch}_1\,\mathrm{ch}_2+2\,\mathrm{ch}_3\,.$$

Example 3.20 confirms the correction. Theorem 3.16 gives $\mathrm{ch}_3(\Lambda) = (B_4/4!)\,(\kappa_3 - \sum_i \psi_i^3 + \delta_{2,0} - 3\delta_{0,1})$ with $B_4/4! = -1/720$. Then $2\,\mathrm{ch}_3$ produces the coefficient $-1/360$ printed for λ_3, while $\mathrm{ch}_3/2!$ would produce $-1/1440$.

The Hodge bundle. Let $\Lambda = \pi_*\omega_\pi$ and $\lambda_j = c_j(\Lambda)$. Its special property, from Theorem 3.16, is $\mathrm{ch}_{2k}(\Lambda) = 0$ for $k \geq 1$. This is a degree count (p. 703): GRR with $\pi_! L = \Lambda - \mathbb{C}$ (Proposition 3.11) gives

$$\mathrm{ch}(\Lambda) - 1 = \pi_*\Big(1 + \frac{\omega}{2} + \sum_{k\geq 1} \Gamma_{2k}\Big), \qquad \Gamma_{2k} = \frac{B_{2k}}{(2k)!}\Big[K^{2k} - \sum_i D_i^{2k} + \Delta \cdot \frac{\nu_1^{2k-1} + \nu_2^{2k-1}}{\nu_1 + \nu_2}\Big] \in H^{4k}(\mathcal{C}).$$

Since π_* lowers degree by two, these terms land only in H^0 and H^{4k-2}, i.e. in ch_0 and the odd ch_{2k-1}. The corrected Proposition 3.17 is therefore

$$\boxed{1 + \lambda_1 + \cdots + \lambda_g = \exp\big(\mathrm{ch}_1(\Lambda) + 2!\ \mathrm{ch}_3(\Lambda) + 4!\ \mathrm{ch}_5(\Lambda) + \cdots\big),}$$

and, for example, $\lambda_2 = \lambda_1^2/2$.

Corollary 3.19: $c(\Lambda)c(\Lambda^\vee) = 1$**.** For any E, the roots of $E^\vee$ are $-x_j$, so $\mathrm{ch}_m(E^\vee) = (-1)^m\,\mathrm{ch}_m(E)$. Adding the exponents for E and $E^\vee$, the odd terms cancel and the even terms double:

$$\boxed{c(E)\,c(E^\vee) = \exp\Big(-2\sum_{k\geq 1}(2k-1)!\ \mathrm{ch}_{2k}(E)\Big).}$$

Because log is injective on $1 +$ (nilpotents) and the summands have distinct degrees, $c(E)c(E^\vee) = 1$ if and only if $\mathrm{ch}_{2k}(E) = 0$ for all $k \geq 1$. For Λ this is Theorem 3.16, which proves the corollary. The property is not formal: for $E = \mathcal{O}_{\mathbb{P}^2}(1)$ we get $(1+h)(1-h) = 1-h^2 \neq 1$, matching $\mathrm{ch}_2(E) = h^2/2 \neq 0$. The degree-4 and degree-$4g$ parts of $(1 + \lambda_1 + \cdots + \lambda_g)(1 - \lambda_1 + \cdots + (-1)^g\lambda_g) = 1$ give

$$2\lambda_2 = \lambda_1^2, \qquad \lambda_g^2 = 0 \quad (g \geq 1).$$

6 Why $\psi_{n+1} = K$ (Lemma 3.21, pp. 705–706)

6.1 The statement

One space, two descriptions. By Proposition 2.2 (pp. 682–683), the forgetful map $p : \overline{\mathcal{M}}_{g,n+1} \to \overline{\mathcal{M}}_{g,n}$ and the universal curve $\pi : \mathcal{C} \to \overline{\mathcal{M}}_{g,n}$ are the same family. Write

$$\Phi : \overline{\mathcal{M}}_{g,n+1} \xrightarrow{\sim} \mathcal{C}, \qquad \pi \circ \Phi = p,$$

for this identification. Each side has its own natural classes, and Lemma 3.21 is a dictionary between them. On $\mathcal{C}$ we use the classes of p. 690:

- $L = \omega_\pi$, the relative cotangent line bundle (Definition 2.6, p. 685). At a smooth point q of a fiber C, its fiber is T_q^*C. Near a node $xy = s$, it is generated by $dx/x = -dy/y$.
- $D_i \subset \mathcal{C}$, the ith marked section, and $D = \sum_i D_i$.
- $L(D)$, which Zvonkine writes $L^{\log}$. Its local sections are relative 1-forms that may have a simple pole along each D_i.
- $K = c_1(L(D)) = c_1(L) + D$; and Δ, the locus of nodes.

A point of $\overline{\mathcal{M}}_{g,n+1}$ is a stable curve $(C'; x_1, \ldots, x_{n+1})$, and on $\overline{\mathcal{M}}_{g,n+1}$ we use:

- L_{n+1}, whose fiber is the cotangent line $T^*_{x_{n+1}}C'$, and $\psi_{n+1} = c_1(L_{n+1})$ (Definition 2.12, p. 686);

- $\delta_{(i,n+1)}$, the divisor of curves with a rational component carrying exactly x_i, x_{n+1} and one node;
- $\delta_{(n+1)}$, the codimension-two locus of curves with a rational component carrying exactly x_{n+1} and two nodes.

Lemma 3.21 (p. 705). Under Φ the following objects correspond.

on $\overline{\mathcal{M}}_{g,n+1}$	on $\mathcal{C}$	codimension
$\delta_{(i,n+1)}$	D_i	1
$\delta_{(n+1)}$	Δ	2
ψ_{n+1}	$K = c_1(L) + D$	—

We will in fact see the stronger statement $L_{n+1} \cong \Phi^*\big(L(D)\big)$ as line bundles.

Why the last line is surprising. A point of $\mathcal{C}$ is a curve $(C; p_1, \dots, p_n)$ together with a point $q \in C$. Usually, the corresponding point of $\overline{\mathcal{M}}_{g,n+1}$ is "the same curve, with q promoted to the marked point x_{n+1}". Then L and L_{n+1} have the same fiber T_q^*C, and one expects $\psi_{n+1} = c_1(L)$. The lemma says that this naive guess is wrong, and the error is exactly the class D of the marked sections.

A degree count shows that the guess must fail. Restrict to the fiber of π over a smooth curve $(C; p_1, \dots, p_n)$. This fiber is C itself, and under Φ it is also the fiber of p. On it, L restricts to ω_C, of degree $2g-2$, while $L(D)$ restricts to $\omega_C(p_1 + \cdots + p_n)$, of degree $2g-2+n$. For $g = 0$, $n = 3$ this fiber is all of $\overline{\mathcal{M}}_{0,4} \cong \mathbb{P}^1$. The guess would give $\int_{\overline{\mathcal{M}}_{0,4}} \psi_4 = -2$, but the correct value is 1 (Section 6.7).

Where the error comes from. "The same curve" is false when q is a marked point p_i or a node. There the $(n+1)$-pointed curve is not C, because a new rational component appears. The rest of this section explains how this *bubbling* changes the cotangent line. Along the marked sections it produces the correction $+D$; along the nodes it produces no correction.

6.2 Three kinds of points

The map Φ (proof of Proposition 2.2, pp. 682–683). Start with $(C'; x_1, \dots, x_{n+1}) \in \overline{\mathcal{M}}_{g,n+1}$. Forget x_{n+1} and stabilize. This gives a stable curve $(\widehat{C}; \hat{x}_1, \dots, \hat{x}_n) \in \overline{\mathcal{M}}_{g,n}$ and a stabilization map $\mathrm{st} : C' \to \widehat{C}$, which contracts the components that have become unstable. Put $y = \mathrm{st}(x_{n+1})$. Then

$$\Phi(C'; x_1, \dots, x_{n+1}) = \big((\widehat{C}; \hat{x}_1, \dots, \hat{x}_n),\, y\big) \in \mathcal{C}.$$

Forgetting x_{n+1} can only destabilize the component R that contains x_{n+1}. This happens when R is rational with exactly three special points, one of which is x_{n+1}. After R is contracted, every other component is still stable. So there are three cases (Figure 1).

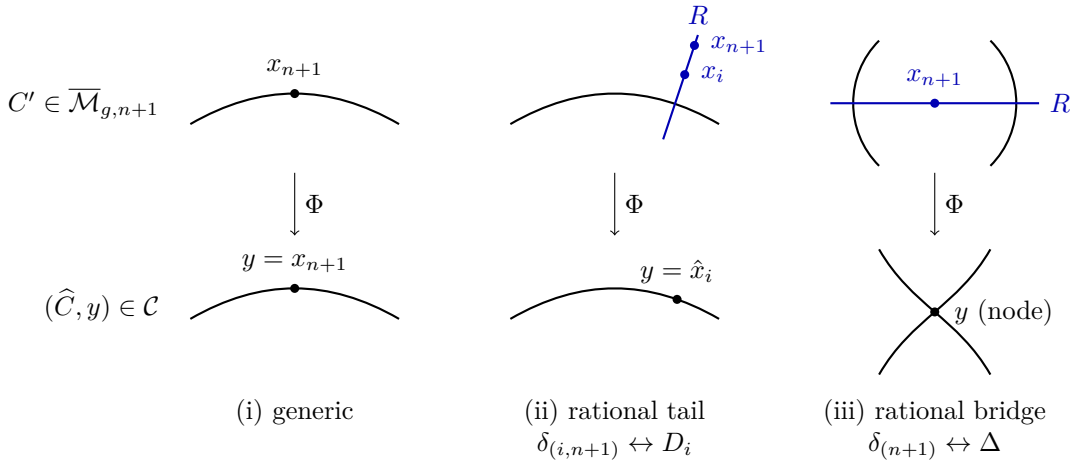
(i) *Generic.* C' is still stable without x_{n+1}. Then st is the identity, and $y = x_{n+1}$ is a smooth, unmarked point.

(ii) *Rational tail.* $R \cong \mathbb{P}^1$ carries x_i, x_{n+1} and one node. Then st contracts R to the point where it was attached, and $y = \hat{x}_i$. So Φ maps $\delta_{(i,n+1)}$ onto D_i.

(iii) *Rational bridge.* $R \cong \mathbb{P}^1$ carries x_{n+1} and two nodes. Then st contracts R, the two branches on either side of R meet, and y is a node of $\widehat{C}$. So Φ maps $\delta_{(n+1)}$ onto Δ.

The inverse map. Given $((C; p_1, \dots, p_n), q) \in \mathcal{C}$, undo the three cases:

- if q is smooth and unmarked, take $C' = C$ and $x_{n+1} = q$;



(i) generic

(ii) rational tail
$\delta_{(i,n+1)} \leftrightarrow D_i$

(iii) rational bridge
$\delta_{(n+1)} \leftrightarrow \Delta$

Figure 1: The three kinds of points of $\overline{\mathcal{M}}_{g,n+1}$ and their images under Φ. Black: the components of $\widehat{C}$; blue: the component R contracted by st. Compare the figure on p. 683.

- if $q = p_i$, glue the point ∞ of a copy of $\mathbb{P}^1$ to p_i, and put $x_i = 0$ and $x_{n+1} = 1$ on it;
- if q is a node, separate its two branches, glue them to the points 0 and ∞ of a copy of $\mathbb{P}^1$, and put $x_{n+1} = 1$.

A copy of $\mathbb{P}^1$ with three special points has no moduli, since a Möbius transformation moves any three points to $0, 1, \infty$. So the new component adds no parameters. This gives the first two lines of the dictionary, and a dimension count confirms the codimensions: $\delta_{(i,n+1)}$ is parametrized by $(C; p_\bullet) = (C; p_1, \dots, p_n) \in \overline{\mathcal{M}}_{g,n}$ alone, so it has dimension $3g - 3 + n = \dim \overline{\mathcal{M}}_{g,n+1} - 1$, like the section $D_i \cong \overline{\mathcal{M}}_{g,n}$. A point of $\delta_{(n+1)}$ is a nodal curve in $\partial\overline{\mathcal{M}}_{g,n}$ together with one of its nodes, so $\delta_{(n+1)}$ has dimension $3g - 4 + n$, like Δ.

6.3 Comparing the two cotangent lines

The comparison map. The stabilization $\mathrm{st} : C' \to \widehat{C}$ is a holomorphic map of curves, so it pulls back cotangent vectors. When y is a smooth point of $\widehat{C}$, i.e. in cases (i) and (ii), this gives a linear map

$$\mathrm{st}^* : \ T^*_y \widehat{C} \longrightarrow T^*_{x_{n+1}} C'.$$

The source is the fiber of L at $\Phi(C')$, and the target is the fiber of L_{n+1} at C'. As C' varies, these maps form a map of line bundles

$$\mathrm{st}^* : \ \Phi^* L \longrightarrow L_{n+1} \qquad \text{on } \overline{\mathcal{M}}_{g,n+1} \setminus \delta_{(n+1)}.$$

The local computation below shows that it is holomorphic. What does it do?

- In case (i), st is the identity near x_{n+1}, so st^* is an isomorphism.
- In case (ii), st collapses the whole tail R to one point. Its derivative at x_{n+1} is zero, so $\mathrm{st}^* = 0$.

So the comparison map is an isomorphism except along the divisors $\delta_{(i,n+1)}$, where it vanishes. To find the resulting twist, we need its order of vanishing there.

Line bundles and zeros of maps. A map $s : A \to B$ of line bundles is the same as a section of $\mathrm{Hom}(A, B) = A^\vee \otimes B$. In local frames a and b, it reads $s(a) = f\, b$ for a holomorphic function f. Suppose s vanishes exactly to first order along a smooth divisor E, i.e. f is a local equation of E times a unit. Then $A^\vee \otimes B \cong \mathcal{O}(E)$, that is,

$$B \cong A(E), \qquad c_1(B) = c_1(A) + [E].$$

The order of vanishing along an irreducible divisor can be read on any small disc crossing E transversally at a general point: it is the order of the zero of f restricted to that disc.

So everything reduces to one claim, checked in the next subsection: *on a disc crossing $\delta_{(i,n+1)}$ transversally,* st* *has a simple zero.* Granting it, and using $\Phi(\delta_{(i,n+1)}) = D_i$, we get

$$L_{n+1} \cong \Phi^* L\Big(\sum_i \delta_{(i,n+1)}\Big) = \Phi^*\big(L(D)\big) \qquad \text{on } \overline{\mathcal{M}}_{g,n+1} \setminus \delta_{(n+1)}.$$

6.4 The local computation: a point colliding with p_i

The disc. Fix a stable curve $(C; p_1, \dots, p_n)$ and a local coordinate z on C with $z(p_i) = 0$. Let the extra point move towards p_i: $q(t)$ is the point with $z = t$, for t in a small disc $B \subset \mathbb{C}$. In $\mathcal{C}$ this is a small disc in the fiber of π over $[C; p_\bullet]$, crossing D_i transversally at $t = 0$. We compute the family of $(n+1)$-pointed curves $\Phi^{-1}(q(t))$.

For $t \neq 0$ nothing happens: the curve is C with $x_{n+1} = q(t)$, which is case (i). At $t = 0$ the points x_{n+1} and x_i would collide, and this is not allowed. To find the stable limit, put all these curves into the surface $S = C \times B \to B$, with the two sections

$$\sigma_i : \ z = 0, \qquad \sigma_{n+1} : \ z = t.$$

They meet only at the point $O = (p_i, 0)$. Let $\widetilde{S} \to S$ be the blow-up of S at O.

The blow-up in coordinates. Near O, the surface S has coordinates (z, t). The blow-up is covered by two charts,

$$\text{chart 1: } (u, t) \ \text{ with } z = tu; \qquad \text{chart 2: } (z, v) \ \text{ with } t = zv,$$

glued by $v = 1/u$. The exceptional curve $E \cong \mathbb{P}^1$ is $\{t = 0\}$ in chart 1, with coordinate u, and $\{z = 0\}$ in chart 2, with coordinate $v = 1/u$.

- *The fiber over $t = 0$.* In chart 1 it is E. In chart 2 it is $\{zv = 0\} = E \cup \{v = 0\}$, and $\{v = 0\}$ is the strict transform of $C \times \{0\}$, a copy of C. So the central fiber is $C \cup E$, with $p_i \in C$ glued to $u = \infty \in E$. The equation $zv = t$ is the standard local smoothing of a node.
- *The sections.* σ_{n+1} $(z = t)$ becomes $u = 1$, and σ_i $(z = 0$, i.e. $tu = 0)$ becomes $u = 0$, after removing E. They are now disjoint, and they meet E at $u = 1$ and $u = 0$.
- Away from O nothing changes, and the other marked points stay on C.

So the stable limit at $t = 0$ is $C \cup E$ with x_i at $u = 0$, x_{n+1} at $u = 1$, and the node at $u = \infty$. This is exactly the rational tail of case (ii). The coordinate $u = z/t$ acts as a magnifying glass. It looks at the part of C at distance about $|t|$ from p_i, rescaled so that the two colliding points stay at distance one. In the limit this rescaled neighbourhood becomes the new component E (Figure 2).

The two cotangent lines along the disc.

- $\Phi^* L$: at time t its fiber is $T^*_{q(t)} C$, framed by dz for every t.
- L_{n+1}: at time t its fiber is the cotangent line to the fiber $\widetilde{S}_t$ at x_{n+1}, i.e. at $u = 1$. In chart 1 the fibers are the lines $t = \text{const}$, and u restricts to a coordinate on each of them, including E at $t = 0$. So du frames L_{n+1} for every t.
- The stabilization map is the blow-down $\widetilde{S} \to S$, $(u, t) \mapsto (z, t) = (tu, t)$. On the fiber at time t the number t is a constant, so

$$\boxed{\text{st}^*(dz) = d(tu) = t\, du.}$$

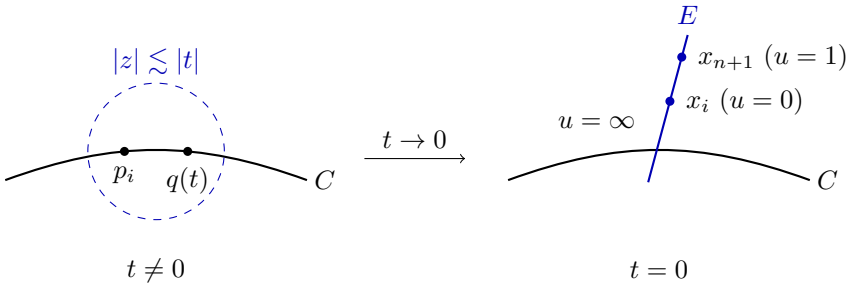


Figure 2: Bubbling. For $t \neq 0$ the point $q(t)$ is at distance $|t|$ from p_i. In the limit, the dashed disc, rescaled by $u = z/t$, becomes the new component E. In the coordinate u, the two points stay at 0 and 1.

In these frames the comparison map is multiplication by t. It is an isomorphism for $t \neq 0$ and zero at $t = 0$, and the zero is simple. The disc crosses $\delta_{(i,n+1)}$ transversally, because it is the preimage under Φ of a disc transverse to D_i. This proves the claim.

This is what Zvonkine means on p. 706: a holomorphic section of L "is naturally extended by 0 on the genus 0 component containing x_i and x_{n+1}", so "the sections of L_{n+1} acquire an additional zero whenever we are in case (ii)". Indeed, pull a 1-form on C back to $C \cup E$. It restricts to zero on E, because E is collapsed to the point p_i: $z = tu$ vanishes identically on E.

Remark (parameters). We kept $(C; p_1, \dots, p_n)$ fixed and moved only q. In families, the coordinate z also depends on the moduli of $(C; p_\bullet)$. The universal curve over $\mathcal{C}$ is then built from $\mathcal{C} \times_{\overline{\mathcal{M}}_{g,n}} \mathcal{C}$ by blowing up the locus where the diagonal meets the ith section (Knudsen's construction). The moduli of $(C; p_\bullet)$ are carried along as passive parameters, and the computation does not change.

6.5 The logarithmic point of view

The same computation fits in one line. On the fibers, $z = tu$ gives

$$\frac{dz}{z} = \frac{d(tu)}{tu} = \frac{du}{u}.$$

The logarithmic form dz/z is invariant under the rescaling $z \mapsto z/t$, and this rescaling is exactly what bubbling does. Now dz/z frames $L(D)$ near D_i, while du/u at $u = 1$ is just du, which frames L_{n+1} for every t, including $t = 0$. So a frame goes to a frame, uniformly in t:

$$\Phi^*\big(L(D)\big) \xrightarrow{\sim} L_{n+1} \quad \text{near } \delta_{(i,n+1)}, \qquad \frac{dz}{z} \longmapsto du.$$

The two computations agree. At the moving point $z = t$, the covector dz/z equals $t^{-1}dz$, and $\mathrm{st}^*(t^{-1}dz) = t^{-1} \cdot t\, du = du$. Dividing by z, i.e. allowing a pole along D_i, exactly cancels the zero created by the bubble.

Moral. L_{n+1} is not the cotangent bundle of the fibers but the *logarithmic* cotangent bundle $\omega_\pi(D)$. Logarithmic forms do not notice bubbling, which is why $\omega_\pi(D)$ is the bundle that survives the identification Φ.

6.6 Case (iii): the nodes do not matter

At a point of $\delta_{(n+1)}$, $y = \mathrm{st}(x_{n+1})$ is a node of $\widehat{C}$, so st^* is not defined there. Zvonkine dismisses this case because $\delta_{(n+1)}$ has codimension two (p. 706). There are two ways to make this precise.

Topology (Zvonkine's argument). Let Z be a closed complex submanifold of complex codimension two, i.e. real codimension four, in a complex manifold X. Then the restriction $H^2(X; \mathbb{Q}) \to$

$H^2(X \setminus Z; \mathbb{Q})$ is injective. Its kernel is the image of $H^2(X, X \setminus Z; \mathbb{Q})$, and by excision and the Thom isomorphism $H^k(X, X \setminus Z) \cong H^{k-4}(Z) = 0$ for $k < 4$. Apply this to $X = \overline{\mathcal{M}}_{g,n+1}$, $Z = \delta_{(n+1)}$ (which is smooth, like Δ: locally $x = y = 0$ inside $xy = s$), and the line bundle $L_{n+1} \otimes \Phi^* L(D)^\vee$, which is trivial on $X \setminus Z$. Its c_1 restricts to 0 on $X \setminus Z$, so it is 0; hence $\psi_{n+1} = \Phi^* K$. On the stack, the same argument works in orbifold charts with rational coefficients.

Algebra (Hartogs). On a smooth space, an isomorphism of line bundles defined outside a subset of codimension two extends to an isomorphism. In local frames it is a nowhere vanishing holomorphic function on $U \setminus Z$, and by Hartogs' theorem both this function and its inverse extend holomorphically across Z. So in fact $L_{n+1} \cong \Phi^*\big(L(D)\big)$ on all of $\overline{\mathcal{M}}_{g,n+1}$.

A direct look (optional). The logarithmic trick also shows directly that nothing happens at the nodes. Near a node of a fiber, $\mathcal{C}$ has coordinates (x, y), plus passive parameters, with $\pi(x, y) = s = xy$, and $L = \omega_\pi$ is framed by $dx/x = -dy/y$. Let $q = (x_0, y_0)$ with $x_0 y_0 \neq 0$. On the fiber $xy = x_0 y_0$ use the coordinate $u = x/x_0 = y_0/y$, so that q is at $u = 1$. As $(x_0, y_0) \to (0, 0)$, this coordinate survives on the limiting bridge of case (iii), with its nodes at $u = 0$ and $u = \infty$. And

$$\frac{dx}{x} = \frac{du}{u} = du \quad \text{at } u = 1.$$

So the frame dx/x of L goes to the frame du of L_{n+1}, uniformly up to the bridge. There is no D near the nodes and no correction is needed: ω_π is already logarithmic at the nodes.

6.7 Checks and an example

Restriction to $\delta_{(i,n+1)}$. By Proposition 2.23 (p. 691), $K \cdot D_i = 0$, because the residue trivializes $L(D)|_{D_i}$. On the other side, the tail $(R; x_i, x_{n+1}, \text{node})$ is isomorphic to $(\mathbb{P}^1; 0, 1, \infty)$ through a *unique* coordinate u. So the cotangent line at x_{n+1} is canonically framed by du, and L_{n+1} is trivial on $\delta_{(i,n+1)}$. The two trivializations match: dz/z has residue 1 along D_i and corresponds to du. Hence $\psi_{n+1} \cdot \delta_{(i,n+1)} = 0$, as the lemma predicts.

Degree on a fiber, and the dilaton equation. On the fiber of p over a smooth $(C; p_1, \ldots, p_n)$, the class ψ_{n+1} has degree $\deg \omega_C(p_1 + \cdots + p_n) = 2g - 2 + n$. Globally, $p_* \psi_{n+1} = \pi_* K = \kappa_0 = 2g - 2 + n$ (p. 692). Each marked point contributes $+1$, one for each bubble. Combined with the previous check and Proposition 2.17, $\psi_i = p^* \psi_i + \delta_{(i,n+1)}$, this gives $\psi_{n+1} \psi_i^a = \psi_{n+1}\, p^* \psi_i^a$, and the projection formula yields the dilaton equation

$$\int_{\overline{\mathcal{M}}_{g,n+1}} \psi_{n+1} \prod_{i=1}^{n} \psi_i^{a_i} = (2g - 2 + n) \int_{\overline{\mathcal{M}}_{g,n}} \prod_{i=1}^{n} \psi_i^{a_i}.$$

Example: $\overline{\mathcal{M}}_{0,4}$. Here $\overline{\mathcal{M}}_{0,3}$ is a point, the curve $(\mathbb{P}^1; 0, 1, \infty)$, and its universal curve is this $\mathbb{P}^1$. So Φ identifies $\overline{\mathcal{M}}_{0,4}$ with $\mathbb{P}^1$: a 4-pointed curve goes to the position q of x_4 in the coordinate z that puts x_1, x_2, x_3 at $0, 1, \infty$. The three points $q = 0, 1, \infty$ are the boundary points $\delta_{(1,4)}$, $\delta_{(2,4)}$, $\delta_{(3,4)}$, where x_4 has collided with x_i and bubbled off with it. The fiber is smooth, so case (iii) does not occur. The lemma predicts $\int_{\overline{\mathcal{M}}_{0,4}} \psi_4 = \deg \omega_{\mathbb{P}^1}(0 + 1 + \infty) = -2 + 3 = 1$. We check this by counting the zeros and poles of explicit sections of L_4, using only the chart of Section 6.4.

(a) A logarithmic form. Let $\alpha = dz/(z(z-1))$, a section of $L(D)$. For $q \notin \{0, 1, \infty\}$, the value $\alpha(q) \in T_q^* \mathbb{P}^1$ lies in the fiber of L_4. This defines a section s_α of L_4 away from the boundary.

- Near $q = 0$, with $u = z/q$, $\alpha = \frac{1}{z-1}\frac{dz}{z} = \frac{1}{qu-1}\frac{du}{u}$. At x_4 ($u = 1$) this is $\frac{1}{q-1} du$, which tends to $-du \neq 0$. So s_α has no zero at $q = 0$, and likewise none at $q = 1$.
- Near $q = \infty$, use $w = 1/z$, $w_0 = 1/q$ and $u = w/w_0$. Then $\alpha = -\frac{dw}{1-w} = -\frac{w_0 u}{1 - w_0 u}\frac{du}{u}$, which at $u = 1$ is $-\frac{w_0}{1-w_0} du$: a simple zero at $w_0 = 0$.

So s_α has exactly one simple zero, and $\deg L_4 = 1$.

(b) A holomorphic form. Now take dz, a section of L with a double pole at ∞. The naive guess would give degree -2. Bubbling changes the count.

- Near $q = 0$: $\mathrm{st}^* dz = q\,du$, a simple zero. Near $q = 1$, the same computation with $z - 1$ in place of z gives another simple zero.
- Near $q = \infty$: $dz = -dw/w^2 = -du/(w_0 u^2)$, which at $u = 1$ is $-w_0^{-1} du$: a simple pole instead of a double pole.

The total is $2-1 = 1$ again. Each of the three marked points adds $+1$ to the naive degree: $-2+3 = 1$.

6.8 The same picture gives Proposition 2.17

Now look at the other marked point on the tail. Along the disc of Section 6.4 the forgetful map p is constant. So $p^* L_i$ has fiber $T^*_{p_i} C$, framed by dz at $z = 0$, while L_i has fiber the cotangent line at x_i, i.e. at $u = 0$, framed by du. Again $\mathrm{st}^*(dz) = t\,du$. So the natural map $p^* L_i \to L_i$ vanishes simply along $\delta_{(i,n+1)}$, and

$$L_i \cong p^* L_i\big(\delta_{(i,n+1)}\big), \qquad \psi_i = p^* \psi_i + \delta_{(i,n+1)},$$

which is Proposition 2.17 (p. 688). Here the logarithmic trick is not available, because du/u has its pole at x_i itself. So the correction remains a boundary divisor on $\overline{\mathcal{M}}_{g,n+1}$.

6.9 Summary and uses

$$\boxed{L_{n+1} \cong \Phi^*(\omega_\pi(D)), \qquad \psi_{n+1} = K = c_1(\omega_\pi) + D.}$$

The cotangent line at the moving point is the relative *logarithmic* cotangent line. The twist by D_i appears because, when the moving point hits p_i, a rational tail bubbles off and holomorphic 1-forms vanish on it. There is no twist by Δ: nodes have codimension two, and ω_π is already logarithmic there.

The lemma is used in Theorem 3.22 (p. 706), where (1) is the lemma itself. It also turns the definition $\kappa_m = \pi_*(K^{m+1})$ (p. 692) into $\kappa_m = p_*(\psi_{n+1}^{m+1})$, the Arbarello–Cornalba definition. In particular $\kappa_0 = 2g - 2 + n$.